

Technical Portfolio

Engineering, Embedded Systems & Software Projects

[YOUR NAME]

Assiut STEM School — Expected Graduation 2027

[EMAIL] | [PHONE]

GitHub: [GITHUB] | LinkedIn: [LINKEDIN] | Portfolio: [PORTFOLIO]

1. Personal Profile

I am a STEM student at Assiut STEM School with a strong focus on embedded systems, robotics, automation, and full-stack software development. Over the past three years I have completed twelve substantial technical projects: five individual embedded-systems and software projects through the Digital Egypt Cubs Initiative, four engineering capstone projects as part of school teams, and three independent freelance software systems built entirely on my own. Across these projects my work has spanned circuit design and prototyping, microcontroller programming, sensor calibration, control-system logic, and serverless web-application architecture, consistently grounded in research, iterative testing, and measurable results.

I did not prioritize heavy competition participation during this period, choosing instead to focus on my academic studies while building depth across this project portfolio.

2. Education

School: Assiut STEM School

Expected Graduation: 2027

GPA: 3.5 (Grades 10 and 11)

Grade 12 GPA: [To be added]

3. Technical Skills

Programming Languages: Python, C/C++, MicroPython, HTML/CSS/JavaScript

Embedded Platforms: ESP32, Raspberry Pi Pico, Arduino

Simulation & Design Tools: Wokwi, CoppeliaSim, Tinkercad

Web / Backend: Firebase (Firestore, Realtime Database, Authentication, Storage)

Version Control: Git, GitHub

4. Digital Egypt Cubs Initiative

The Digital Egypt Cubs Initiative is a nanodegree-style technology program sponsored by Egypt's Ministry of Communications and Information Technology (MCIT) and managed by Udacity. Over three years of participation, I completed five individual projects — each carried out independently from initial design through implementation, debugging, and final testing — spanning software development, digital logic, and embedded systems and robotics. My participation progressed through verified program levels, culminating in a Level Three: Embedded Systems and Robotics certificate.

Project 1: Adventure Game

Adventure Game

Individual Project — Digital Egypt Cubs Initiative

Objective: Build a text-based adventure game demonstrating core Python programming concepts, including branching logic and state management.

My Role: Sole developer — designed, implemented, tested, and debugged the entire program independently.

Technologies: Python

Implementation:

- Built a console-based game where the player makes choices that affect story outcomes, using `input()`, `if/else` branching, and `while` loops.
- Used the `random` module to introduce random events and `time.sleep()` to pace narrative delivery.
- Tracked game state using variables such as `total_score`, and structured the program into reusable functions with comments throughout.
- Implemented input validation loops to reject invalid user entries before proceeding.

Key Concepts: Control flow (`if/else`, `while`), functions, state management, input validation, replayability design, and adherence to PEP 8 style conventions.

Testing: Manually tested each branching path and win/loss condition, and validated that invalid inputs were correctly rejected and re-prompted.

Results: A fully playable, replayable text-based adventure game with multiple distinct endings and randomized events.

Project 2: Digital Device-Control Interface

Digital Device-Control Interface

Individual Project — Digital Egypt Cubs Initiative

Objective: Design a microprocessor-based interface that allows a user to control devices only after entering a correct passcode, combining digital logic theory with embedded programming.

My Role: Sole developer — designed the digital logic, built the circuit, and implemented the control program independently.

Technologies: MicroPython, Raspberry Pi Pico, Wokwi

Key Concepts: Digital logic (AND, OR, NOT gates), multiplexing, binary representation and calculation, decoding, MCU input/output pins, interrupts, and polling.

Implementation:

- Built the circuit in the Wokwi simulator, wiring pushbuttons, LEDs, and a multiplexer to the microcontroller.
- Implemented passcode-checking control logic on the microcontroller, translating button input sequences into binary comparisons.
- Worked within embedded resource constraints (limited I/O pins and processing power) typical of microcontroller-based systems.

Challenges: Mapping digital logic gate behavior (AND/OR/NOT) onto a practical multiplexed circuit and reconciling it with the microcontroller's available GPIO pins.

Results: A working passcode-gated device-control interface demonstrating correct digital logic evaluation and I/O handling on the Raspberry Pi Pico.

Project 3: Embedded Voltmeter

Embedded Voltmeter

Individual Project — Digital Egypt Cubs Initiative

Objective: Build an embedded voltmeter that reads an analog input, converts it to a voltage value, and displays the result on a physical display.

My Role: Sole developer — designed the circuit and wrote all firmware independently.

Technologies: Raspberry Pi Pico, MicroPython, Wokwi, ADC, four-digit seven-segment display

Implementation:

- Read an analog signal via the Pico's ADC and converted the raw reading into a voltage value within the 0–3.3V range.
- Applied averaging to reduce reading fluctuations and implemented button debouncing for reliable user input.
- Used an edge-triggered interrupt and timer-based display scanning to drive the four-digit seven-segment display without blocking other logic.
- Applied precise rounding to the displayed voltage values.
- Designed the circuit and mapped breadboard connections to the correct MCU pins, then debugged and tested the full system.

Key Concepts: Analog-to-digital conversion, signal averaging/filtering, interrupt-driven I/O, and multiplexed display scanning.

Results: A functioning embedded voltmeter accurately displaying analog voltage readings on a seven-segment display.

Project 4: Smart Parking System

Smart Parking System

Individual Project — Digital Egypt Cubs Initiative

Objective: Build a subsystem that automatically controls a parking entrance barrier based on vehicle detection and space availability.

My Role: Sole developer — designed and implemented the complete control subsystem independently.

Hardware: ESP32, HC-SR04 ultrasonic sensor, servo motor

Implementation:

- Took distance readings from the ultrasonic sensor every 200 ms, discarding readings of 0 or greater than 400 cm as invalid.
- Required 2 consecutive valid readings at or below 100 cm before classifying the object as an arriving vehicle, reducing false triggers.
- Opened the barrier automatically when a parking space was available, and closed it after the vehicle entered or after a 30-second timeout.
- Sent status information over UART to report system state.

Engineering Constraint: The system was required to use non-blocking, loop-based logic for all sampling, timing, debounce, and timeout behavior — explicitly without `Pin.irq` or `Timer` callbacks — which required careful state-machine-style design within the main loop.

Results: A working non-blocking barrier-control subsystem that reliably distinguished valid vehicle detections from sensor noise and correctly managed barrier open/close timing.

Project 5: Smart Delivery System

Smart Delivery System

Individual Project — Digital Egypt Cubs Initiative

Objective: Build a two-part smart delivery system consisting of an access-controlled control station and an autonomous delivery robot, sharing a common scenario and terminology without direct communication between the two subsystems.

My Role: Sole developer — designed and implemented both the control-station firmware and the robot simulation independently.

Technologies — Control Station: ESP32, MicroPython, Wokwi, MQTT, microSD logging

Technologies — Delivery Robot: CoppeliaSim, Python, Pioneer3DX

Control Station Implementation:

- Implemented access control via a user-entered code, with a lockout of 30 seconds triggered after 3 failed attempts.
- Built destination selection, status indicators, and a non-blocking 40-second delivery timer running from 0 to 100.
- Integrated MQTT messaging and microSD logging for status/data persistence.

Delivery Robot Implementation:

- Simulated an autonomous Pioneer3DX robot in CoppeliaSim navigating a scene containing HOME, Destination A, B, and C, walls, and fixed obstacles.

- Implemented obstacle avoidance using proximity sensors, destination detection, position tracking, and distance calculations, with console output for monitoring.
- Met the rubric requirement that the robot stop within 0.20 m of its target destination.

Results: A functioning two-part system: an access-controlled delivery station with reliable lockout/timer logic, and an autonomous simulated robot that navigated to destinations while avoiding obstacles and stopping within the required 0.20 m tolerance.

5. Assiut STEM School Engineering Projects

The following four projects were completed as part of school engineering teams at Assiut STEM School. On each team, my role centered on writing and documentation, implementation, and prototype development, with significant contributions to research and identifying design solutions. These were team efforts, and the descriptions below reflect my individual contributions within each group project rather than sole ownership of the work.

Project A: Hydraulic Multi-Functional Bridge

Hydraulic Multi-Functional Bridge

School Team Project

Project Type: School Team Project

Objective: Design a multi-functional bridge to help reduce traffic congestion by providing dedicated space for pedestrians and cyclists, while remaining capable of supporting heavy vehicle loads.

Problem: Congested crossings often force pedestrians, cyclists, and heavy vehicle traffic to compete for the same limited space, and fixed-span bridges cannot accommodate boat traffic passing beneath them.

My Role: Contributed to the bridge's design and researched candidate solutions; worked with palm fronds/jreed as a construction material for the prototype; performed load and force calculations; worked on the hydraulically actuated movable section; and contributed to building the physical prototype.

Key Feature: A movable section of the bridge, actuated hydraulically, allowed boats to pass underneath when needed. This movable section was functional in the completed prototype.

Materials: Palm fronds/jreed were used as a construction material for the model, combined with a hydraulic system for the movable span.

Engineering Concepts: Load and force calculations for structural members under vehicle loading; hydraulic actuation for the movable span.

Results: A working scale prototype combining a fixed load-bearing span for vehicles with pedestrian/cyclist space, and a functional hydraulically operated movable section for boat clearance.

Project B: Solar Powered Pumped Hydro Storage (PHS) System

Solar Powered Pumped Hydro Storage

Grade 10, Semester 2, 2024/2025 — Group 14128

Team: Ahmed Mohamed Abdallah Mostafa, Ahmed Abdel Latif Mahmoud Mohamed, Gamil Mamdouh El Gamil Asham, Moaaz Abu Bakr Anwar Mohamed

Objective: Design a small-scale system that stores solar energy by pumping water to an elevated reservoir, then recovers that energy as electricity by releasing the water through a turbine generator — addressing Egypt's need for alternative, non-fossil-fuel energy sources.

Problem: Egypt faces pressure to reduce dependence on non-renewable energy sources and their associated pollution, while needing cost-effective, small-scale, environmentally friendly energy-storage solutions suitable for education and remote-area use.

My Role: Contributed to research, design, prototype construction, and testing as part of the team; the project used recycled materials (including a recycled barrel and a solar panel recycled from an earlier project) to reduce cost and environmental impact.

Hardware / Materials: 18V 10W solar panel, 12V DC water pump, water turbine generator, lithium batteries, hose, recycled barrel, DC-DC converter/regulator

Software / Design Tools: Tinkercad (used for the initial 3D design of the prototype)

Design Requirements:

- Produce more than 40 J of energy.
- Reach that output within 5 minutes or less.
- Keep total prototype volume under 0.5 m³.
- Power the water pump using solar energy.

Implementation: The solar panel charges a lithium battery system that powers a 12V DC pump, lifting water from a lower reservoir (simulated by a barrel) to an upper reservoir. As water falls back down, it passes through a water turbine generator, and voltage/current were measured with a digital multimeter (Avometer) to calculate power ($P = IV$) and energy ($E = PT$).

Testing & Iteration:

- Trial 1: Water leakage from the barrel and low water velocity produced only ~7.7 J over 5 minutes — requirements not met.
- Trial 2: Leakage was fixed and tap area reduced to increase water velocity, raising output to ~23 J over 5 minutes — still short of the target.
- Trial 3: A larger-volume barrel increased the height/pressure difference driving flow, achieving the best result.

Results: The final trial produced approximately 87.1 J in 5 minutes, and reached the 40 J requirement in roughly 2.5 minutes, while keeping the prototype's total volume (~0.375 m³) under the 0.5 m³ limit.

Improvements / Future Work:

- Add IoT monitoring for real-time water flow, charge level, and turbine performance.
- Upgrade to a higher-efficiency water pump, generator, batteries, and solar panel.
- Reduce friction losses within the piping to improve flow efficiency.

Project C: Rollmetric

Rollmetric — Digital Measuring Wheel & Inclinometer

Grade 11, Semester 1, 2025/2026 — Code No. 14228

Team: Mohamed Alaa, Mohamed Ali, Gamil Mamdouh, Youssef Mostafa

Objective: Build a dual-purpose device — a digital measuring wheel and an inclinometer — from components extracted from e-waste, achieving distance and angle measurement errors below ±5%.

Problem: Electronic waste is a growing source of air, water, and soil pollution; the project set out to convert discarded electronic components into a precise, functional, and sustainable measuring instrument rather than adding to landfill waste.

My Role: Contributed to design, prototype construction, and testing as part of the team.

Hardware: ESP32 (LilyGo TTGO T-Display), two Hall-effect sensor modules, MPU6050 accelerometer, Neodymium magnets, Li-ion battery, acrylic enclosure, wooden wheel

Implementation: A wheel fitted with evenly spaced magnets rotates as the device is pushed; two Hall-effect sensors detect each magnet passing and, using the order in which the two sensors trigger, determine both distance traveled and direction of rotation. The MPU6050 accelerometer, mounted level inside the enclosure using laser levels for alignment, measures surface inclination via its Z-axis gravitational component.

Key Development — Magnet Spacing:

- Magnet spacing was calculated as $R \times 2\pi / n$, where R is the wheel radius and n the number of magnets.
- An initial 5-magnet configuration produced incorrect spacing (~1.5 cm) and a distance error of 33.3%.
- The design was corrected to 10 magnets, achieving a spacing of 0.999 ± 0.0004 cm and reducing distance error dramatically.

Key Development — Inclination Measurement:

- An early mechanical inclinometer (ball motion under gravity) suffered friction and rust, producing angle errors up to 75.4%.
- Replacing it with the MPU6050 3-axis accelerometer improved angle accuracy to 95.8%, and further tuning brought the final angle error down to 4.2%.

Testing: Five iterative trials were conducted, progressively improving distance accuracy from an 18% baseline error down to 0.2%, and angle accuracy from a 75.4% mechanical-inclinometer error down to 4.2% using the accelerometer. Final test runs over paths of ~100 cm measured 102, 98, 100, 99, and 100 cm.

Results: Final distance error of 0.2% and angle error of 4.2%, meeting the project's ±5% design requirement for both measurements.

Engineering Concepts: Hall-effect magnetic field sensing, capacitive accelerometer physics and vector decomposition of gravitational acceleration, and rate-of-change analysis ($m = (y_2 - y_1)/(x_2 - x_1)$) applied to sensor calibration graphs.

Improvements / Future Work:

- Use higher-precision magnetic encoders to further improve distance measurement.
- Add a higher-accuracy accelerometer (e.g., ADXL355) for improved angle measurement.
- Add wireless (Bluetooth/Wi-Fi) data transfer for remote data collection.

Project D: Multi-Parameter Feedback Control for Automated Greenhouse Systems

Automated Greenhouse Control System

Grade 11, Semester 2, 2025/2026 — Group No. 14229

Team: Gamil Mamdouh, Mahmoud Mohamed, Ahmed Abdullatif, Abanob George

Objective: Design a low-cost, closed-loop feedback control system for greenhouse agriculture that autonomously monitors and regulates soil moisture, air humidity, and irrigation water salinity for multiple crop profiles.

Problem: Egypt's agricultural sector faces water scarcity, soil salinization, and arid conditions, particularly in Upper Egypt; manual monitoring of greenhouse conditions is error-prone and wastes water and resources.

My Role: Contributed to research, design, prototype construction (built primarily from recycled plastic containers and reclaimed framing), and testing as part of the team.

Hardware: ESP32 microcontroller, capacitive soil moisture sensor, DHT22 humidity/temperature sensor, analog TDS sensor, 12V DC water pump (1.2 L/min), solenoid valve, dual 5V cooling fans, 4-channel relay module, buzzer

Implementation: A rule-based IF-THEN control algorithm samples all sensors every 10 seconds with a 2-minute hysteresis window. The system activates irrigation when soil moisture falls below a crop-specific threshold (provided salinity is within range), engages cooling fans when humidity or temperature exceeds the upper setpoint, and inhibits irrigation with a buzzer alert if salinity reaches the maximum threshold. Two switchable crop profiles (Basil and Okra) each define their own target ranges for moisture, TDS, humidity, and temperature.

Testing & Iteration:

- Trial 1 (Basil): Soil moisture sensor readings spiked to 88.7% during spray cycles due to the pump spraying water directly onto the sensor's capacitive plate; the sensor was repositioned to fix this.
- Trial 2 (Basil): A humidity control event saw humidity rise from 63.5% to 75.3% before the fans activated and restored it to 63.9% within 30 seconds — a recovery rate about 22× faster than the accumulation rate.
- Trial 3 (Basil): TDS at 1724 ppm (above the 1600 ppm maximum) was diluted via the solenoid valve down to 1487 ppm within 20 minutes 21 seconds, a reduction rate of about 11.65 ppm/min.
- Trial 4 (Okra): The system was switched to the Okra crop profile in real time, validating that the same hardware could support multiple crop types without reprogramming.
- Trial 5 (Basil): Soil moisture recovered from 34.8% to 63.5% over 16 minutes following an irrigation event.

Results: Across five controlled deviation trials, the system operated autonomously for 6+ hours per session and detected threshold breaches within ≤ 5 seconds. Key results included humidity dropping from 84.3% to 73.2% in ~ 4 minutes, TDS diluting from 1748 to 1487 ppm in ~ 15 minutes, and soil moisture recovering from 34.8% to 63.5% in ~ 16 minutes, with combined peak actuator power of 8.5 W (~ 1.10 Wh/session). The prototype met all 12 design requirements of the capstone challenge, including data logging and a wireless web interface for visualization.

Engineering Concepts: Ohm's Law ($V = IR$) for circuit safety and current/voltage sizing; the Work/Power/Efficiency relationship ($P = W/T$) for actuator selection; rate-of-change modeling ($m = (y_2 - y_1)/(x_2 - x_1)$) for settling-time analysis of sensor data; and feedback control theory governing sensor–actuator interaction.

Improvements / Future Work:

- Upgrade to industrial-grade sensors (e.g., SHT4x humidity sensor) for improved corrosion resistance and long-term accuracy.
- Replace simple IF-THEN logic with PID or fuzzy-logic control for smoother environmental adjustments.
- Add an ESP32-S3 module for cloud integration and remote monitoring via a mobile dashboard.
- Integrate a LiFePO4 battery with an MPPT solar charge controller for continuous day/night operation.

6. Independent / Freelance Projects

The following three systems were designed, built, and deployed entirely on my own as independent freelance work, covering the full stack from data architecture and security to user-facing interface design.

Project 1: Thyatira Team — Entertainment Streaming Platform

Entertainment Streaming Platform

Individual Freelance Project

Objective: Build a Netflix-style streaming platform for a theater group's recorded productions, enabling remote viewing with access control, without relying on paid hosting or a traditional backend server.

Problem: The theater group needed to distribute recorded productions to a wide, remote audience with controlled access, but without the cost and complexity of a dedicated media server or paid hosting infrastructure.

My Role: Sole developer — designed and built the full system independently.

Architecture: Two self-contained HTML files: a public viewer (index.html) and an admin dashboard (dashboard.html), backed entirely by Firebase Firestore, Firebase Authentication, and Firebase Security Rules. Video content is served via YouTube and Google Drive rather than a custom media server.

Technologies: Firebase Firestore, Firebase Authentication (Anonymous for viewers, Email/Password for admins), Firebase Security Rules, HTML/JavaScript

Public Viewer Features:

- Browse seasons, enter an access code and name, and unlock and watch productions.
- Switch between Arabic and English display language.

Admin Dashboard Features:

- Create and manage seasons and productions.
- Generate access codes individually or in bulk, with configurable expiry, usage limits, and device limits.
- Search, copy, extend, disable/revoke, and bulk-revoke codes; manage active sessions; and view usage analytics.

Security / Anti-Piracy Measures:

- A moving viewer/timestamp watermark overlaid on video playback.
- Detection of tab loss, DevTools usage, and screenshot-shortcut attempts.
- Right-click blocking, drag blocking, keyboard-shortcut restrictions, and iframe sandboxing.
- Access codes were used as Firestore document IDs to reduce the risk of enumeration attacks.

Notes: These mechanisms are explicitly deterrents against casual piracy rather than true DRM, and the system does not claim to prevent copying absolutely.

Results: A fully functional, self-hosted streaming platform supporting bulk access-code generation, usage analytics, and multiple deterrence mechanisms, running without any paid backend infrastructure.

Project 2: St. Mark's Church Hymn Ministry Management System

Hymn Ministry Management System

Individual Freelance Project

Objective: Build a complete management system for a church hymn ministry, covering student records, attendance, meetings, exams, grading, and leaderboards.

Problem: The ministry needed a single system to manage student records, track attendance across weekly meetings and Mass, follow up with parents, schedule meetings, and administer exams and grading — previously handled manually.

My Role: Sole developer — designed and built the entire system independently, then performed a follow-up audit to fix identified issues.

Architecture: A single self-contained HTML file using vanilla JavaScript, Firebase Realtime Database, Firebase Storage, sessionStorage, SheetJS (for Excel export), and the browser's native print function. The interface is built right-to-left (RTL) in Arabic with a custom church-themed design.

Student Management:

- Records student name, school year, hymn level, parent contact, points, streak, hobbies/talents, photo, and notes, with search and filtering.

Attendance & Meetings:

- Tracks weekly meeting attendance, Mass attendance, and absentees, with parent follow-up and quick batch attendance entry.
- Supports creating and scheduling recurring meetings on a calendar, with goal tracking.

Hymn Levels, Exams & Leaderboards:

- Students are organized across 8 hymn skill levels, each with its own attendance and points statistics.
- A full exam/gradebook module supports percentage calculation, letter grades, and pass/fail status, with Excel export.
- A leaderboard and servant-role management system, with PDF and Excel export of results.

Post-Launch Audit Fixes:

- Resolved a startup race condition and an XSS gap identified during review.
- Added handling for previously unhandled Firebase errors and fixed a timer leak.
- Eliminated unnecessary full-table re-rendering to improve performance.

Results: A production Arabic-language ministry management system in active use, covering the full workflow from student registration through attendance, grading, and leaderboard reporting.

Project 3: Online Exam / Quiz System

Online Exam / Quiz System

Individual Freelance Project

Objective: Build a serverless system for conducting remote exams and quizzes with automatic grading and live administrative control, without requiring a custom backend server.

Problem: Remote examination requires reliable access control, varied question formats, integrity monitoring, and administrative oversight — typically requiring a dedicated backend server, which this system was designed to avoid.

My Role: Sole developer — designed and built the complete system independently.

Architecture: Two static HTML files — a student exam client and an admin dashboard — backed by Firebase Realtime Database, with no Node.js server, PHP, or custom backend required.

Student Exam Experience:

- Students log in using a one-time access code and their name.
- Supports 5 question types: Multiple Choice, Multi-select, True/False, Matching, and Essay.
- Uses per-question timers alongside a master exam timer.

Exam Monitoring & Integrity:

- Detects tab switching, window blur, window resize, DevTools signals, right-click, and restricted keyboard shortcuts, logging all violations.
- Depending on configuration, can force-submit the exam or assign a zero score after cheating detection.
- Optional webcam monitoring via periodic snapshots.

Admin Dashboard:

- Manage a question bank: create, edit, and delete questions, including uploading question images.
- Generate access codes individually or in bulk, with export, reset, and bulk-management options.
- View and sort results, export to CSV, review individual answers, and manually override scores.
- Control exam settings live, including a kill switch to halt an exam in progress.

Results: A fully serverless exam platform supporting five question types, real-time integrity monitoring, and complete administrative control over question banks, access codes, and live exam sessions.

7. Leadership & School Activities

- Co-organized Computer Science sessions at school with a friend, and delivered Computer Science presentations to peers.
- Helped organize school activities around Ramadan, including contributing money annually toward Ramadan decorations.
- Participated in multiple school Fun Days.
- Participated in the school football league.

I intentionally limited my participation in formal competitions during this period in order to prioritize my academic studies, while still remaining active in school life and technical project work.

8. Certifications

I currently hold 37 certificates from online courses and training programs, spanning programming, AI, data, cloud computing, embedded systems, and professional/soft skills. The table below lists all 37 as extracted from the certificate files. Where two certificate images shared the same certificate/verification code (noted below), they represent a single certificate rendered across more than one page rather than two separate awards.

#	Certificate / Course	Issuing Organization	Date
1	Cloud Basics: Development and Basic Concepts (CRA Training Program)	Huawei ICT Academy	Dec 20, 2025
2	SQL Intermediate	Sololearn	Oct 28, 2025
3	Introduction to SQL	Sololearn	Oct 28, 2025
4	Digital Egypt Cubs Initiative — Level Two (Nanodegree Program)	Udacity / Egypt MCIT	Oct 2024
5–6	(Duplicate images of certificate #4, same verification code ELE6CW2H)	—	—
7	Fundamentals of Digital Marketing (Maharat min Google)	Google	Jul 25, 2024
8	Python Developer	Sololearn	Feb 21, 2025
9	AI for Beginners	HP LIFE	—
10	AI Basic: Overview of AI (CRA Training Program)	Huawei ICT Academy	Dec 21, 2025
11	Artificial Intelligence and Applications	Huawei ICT Academy	Dec 16, 2025
12	Artificial Intelligence Technology and Applications (same code as #11)	Huawei ICT Academy	Dec 16, 2025
13	Digital Egypt Cubs Initiative — Level Three: Embedded Systems and Robotics	Udacity / Egypt MCIT	Oct 2025
14	(Duplicate image of certificate #13, same verification code XTTV9UL6)	—	—
15	Microprocessors and Microcontrollers (Professional Certificate Program)	Academy Europe Open University	Dec 16, 2025
16	Data Management and Analysis (CRA Training Program)	Huawei ICT Academy	Dec 16, 2025
17	Data Management and Analytics (same code as #16)	Huawei ICT Academy	Dec 16, 2025
18	Python Machine Learning: From Beginner to Pro (5.5 hrs)	Udemy (Learnify IT)	Mar 16, 2025
19	Hands-On Python Machine Learning with Real World Projects (4 hrs)	Udemy (Sayman Creative Institute)	Mar 16, 2025
20	The Best ChatGPT & AI Course: Make Money With AI (2 hrs)	Udemy (Gerard Yadav)	Mar 16, 2025
21	HTML 5 With Quizzes And Python 3 Complete Course (7 hrs)	Udemy (Horizon Tech)	Mar 16, 2025
22	Python for Everyone: Master the Basics of Programming (6.5 hrs)	Udemy (Sayman Creative Institute)	Jun 2, 2025
23	Development and Basic Concepts of Cloud Computing	Huawei ICT Academy	Dec 20, 2025
24	Effective Leadership	HP LIFE	—
25	Effective Presentations	HP LIFE	Aug 3, 2026
26	Tech for Everyone	Sololearn	Mar 12, 2025
27	Python	Kaggle	Feb 19, 2025
28	Information Representation and Data Organization (CRA Training Program)	Huawei ICT Academy	Dec 27, 2025
29	(Duplicate image of certificate #28, same code)	—	—
30	Introduction to Modern AI	Cisco Networking Academy	Aug 31, 2026
31	Overview of AI	Huawei ICT Academy	Dec 21, 2025

#	Certificate / Course	Issuing Organization	Date
32	Python Programming Basics	Huawei ICT Academy	Dec 15, 2025
33	Collaborative Artificial Intelligence — A Foundation Course (1 hr)	Udemy (Prince Patni)	Mar 16, 2025
34	Object-Oriented Programming (OOP) — Learn to Code Faster (3.5 hrs)	Udemy (SDE Arts, Ahmed El Mohandes)	Jun 2, 2025
35	HTML5 CSS3 and JavaScript for Beginners: From Zero to Hero (11.5 hrs)	Udemy (Mehmood Khalil)	Mar 16, 2025
36	Numpy For Data Science — Real Time Experience (2.5 hrs)	Udemy (Data Science Lovers)	Mar 16, 2025
37	Python For Data Science — Real Time Experience (4.5 hrs)	Udemy (Data Science Lovers)	Mar 16, 2025

Note: certificates #5–6, #14, and #29 are repeated images of certificates #4, #13, and #28 respectively (identical verification codes), so the number of distinct certificates is 34; all 37 supplied certificate files are listed above for completeness.

9. English Proficiency

Duolingo English Test: Taken; Score: [To be added]

10. Contact / Links

Email: [EMAIL]

Phone: [PHONE]

GitHub: [GITHUB]

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Portfolio Website: [PORTFOLIO]